MLDL project report

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Abstract

This study explores the integration of Federated Learning (FL) and Sparse Training for image classification using Vision Transformers (ViT). FL enables multiple clients to collaboratively train a model without sharing raw data, addressing privacy and scalability concerns in decentralized environments. However, FL introduces challenges such as increased communication costs and reduced performance in the presence of non-IID (heterogeneous) data distributions.

To address these challenges, we combine FL with sparse fine-tuning, where only a subset of model parameters is updated using gradient masking. We evaluate various sparsity strategies—Fisher Information, magnitude-based pruning, random selection, and a hybrid method—within a custom SparseSGDM optimizer.

Experiments are conducted using the CIFAR-100 dataset, partitioned across clients under both IID and non-IID conditions, with heterogeneity controlled by varying the number of classes per client. A ViT-Small model pretrained via DINO is fine-tuned for 100-way classification. Centralized training serves as a performance upper bound, optimized via grid search. Federated training is implemented using the FedAvg algorithm with configurable local epochs and client participation.

Our results show that centralized training yields the highest accuracy (90.3%), while FL suffers under non-IID settings. However, sparse training—particularly using sensitivity-based masks—preserves up to 90% accuracy with 70–90% fewer parameter updates, improving efficiency. A Jaccard-based mask overlap analysis reveals that Fisher-based masks enhance client personalization and reduce drift.

This modular framework supports future extensions and demonstrates that sparse model editing significantly improves FL performance and resource efficiency, offering practical value for real-world decentralized learning systems.

The code is available at `https://github.com/Mohamed-Hatem346906/FL-Project`

1. Introduction

The success of modern deep learning models, particularly in computer vision, hinges on their ability to learn from vast centralized datasets. Vision Transformers (ViTs) [1] have emerged as state-of-the-art architectures for image classification, achieving remarkable performance in centralized training paradigms. However, such paradigms assume unfettered access to raw data, raising critical concerns around privacy, communication costs, and scalability—especially in decentralized environments like edge devices or collaborative networks [2].

Federated Learning (FL) [3] addresses these challenges by enabling collaborative model training across distributed clients without sharing raw data. While FL preserves privacy, it introduces two key bottlenecks:

Communication overhead from frequent parameter exchanges.

Performance degradation under non-IID (non-identically distributed) data, where client-specific data distributions cause "client drift" during aggregation [4].

To mitigate these issues, we propose integrating FL with sparse training, where only a critical subset of model parameters is updated. Sparse training reduces communication and computation costs by transmitting and processing fewer parameters, while theoretically preserving model performance. Prior work has explored sparsity in FL for CNNs [10], but its application to Vision Transformers—whose attention mechanisms may demand different sparsity patterns—remains understudied.

This work systematically investigates sparse training for ViTs in FL, with three key contributions:

- Methodology: We adapt the FedAvg algorithm [3] with a custom SparseSGDM optimizer, leveraging gradient masking based on parameter sensitivity (Fisher Information [9]), magnitude pruning, and hybrid strategies.
- Analysis: We evaluate sparsity-accuracy trade-offs under varying non-IID levels (1–50 classes/client) and introduce a Jaccard-index-based mask overlap metric to quantify client specialization.

- Empirical Insights: We demonstrate that Fisher-based sparse masks retain 90% of centralized accuracy at 70% sparsity while reducing client drift, outperforming heuristic (e.g., magnitude-based) and random baselines.

By bridging FL and sparsity for ViTs, this work advances efficient, privacy-preserving distributed learning. Our modular framework (code available) opens avenues for future extensions, such as dynamic sparsity or adaptive client participation.

2. Related Work

This study is built upon the intersection of two key research areas: Federated Learning (FL) and model editing through sparsity. The foundational work in FL was introduced by McMahan et al. [3] with the Federated Averaging (FedAvg) algorithm, a protocol designed to train models on decentralized data while preserving user privacy. A primary challenge that emerged immediately is the issue of statistical heterogeneity, where clients' local data is not identically and independently distributed (non-IID) [4]. This "client drift" can cause parameter interference during aggregation, leading to unstable convergence. Numerous solutions have been proposed to tackle this, such as FedProx [4], which adds a proximal term to regularize local updates, and SCAFFOLD [6], which uses control variates to correct for client drift. While these methods modify the aggregation algorithm, our work explores a complementary approach by modifying the client-side training itself.

The concept of modifying the training process falls under the umbrella of model editing, where the goal is to intelligently update a model's parameters. Our approach leverages sparse fine-tuning, a technique rooted in the broader field of network pruning [7]. The core idea is that large neural networks are often over-parameterized, and their performance can be maintained or even improved by updating only a small subset of their weights. Foundational methods for identifying which parameters to update (or prune) include heuristics like magnitude-based pruning [8] and more sophisticated techniques based on parameter sensitivity, such as using the Fisher Information to estimate a parameter's importance [9]. Combining sparsity with FL is a promising research direction for improving efficiency and performance [10]. While some have explored this for CNNs, our work contributes by systematically applying and analyzing these sensitivity-based masking strategies on a Vision Transformer (ViT) architecture [1] in a federated context, an area of growing importance [11].

3. Methodology

This section details the technical framework of our study. We describe the baseline model setup, the federated learning implementation, our model editing approach via sparse fine-tuning, and the metrics used for analysis.

3.1. Centralized Baseline Setup

To establish a strong performance benchmark, a centralized baseline was configured.

- Model: We use a Vision Transformer (ViT-S/16) [1], a 16x16 patch-size small variant of the ViT architecture, instantiated via the timm library [12]. The model is initialized with weights pretrained on ImageNet using the DINO self-supervised learning method [5]. The final classification head was replaced with a linear layer for the 100 classes of the target dataset.
- Dataset: The CIFAR-100 dataset [13] was used for all experiments.
- Training: The model was trained using a Stochastic Gradient Descent with Momentum (SGDM) optimizer. The learning rate was managed by a cosine annealing schedule to ensure smooth convergence.

3.2. Federated Learning Implementation

The decentralized experiments were built upon a standard FedAvg protocol with the following specifications.

- Data Partitioning: To simulate different real-world conditions, the CIFAR-100 training data was partitioned among K=100 clients using two schemes:
  1. IID: Data was shuffled and distributed evenly.
  2. Non-IID: Data heterogeneity was controlled by assigning only a subset of classes to each client. We tested several levels of heterogeneity, with the number of classes per client (Nc) set to 1, 5, 10, 50.
- FedAvg Protocol: In each communication round, a fraction C=0.1 (10 clients) was randomly selected. These clients would perform local training for a set number of local steps, J, on their data partition. We experimented with J values of 4, 8, 16 to analyze the impact of local computation on convergence. The server then aggregates the updated models via weighted averaging [3].

3.3. Model Editing via Sparse Fine-Tuning

Our core contribution lies in applying sparse fine-tuning as a model editing technique to mitigate the challenges of FL.

- Sparse Optimizer (SparseSGDM): We implemented a custom optimizer, SparseSGDM. This optimizer functions similarly to a standard SGDM optimizer but incorporates a binary mask, M. At each optimization step, the computed gradient, g, is multiplied element-wise by the mask

$$g_{\text{masked}} = g \cdot M$$

before being applied to the model's weights. This ensures that only parameters corresponding to a 1 in the mask are updated, effectively "freezing" the rest.

- Gradient Mask Calibration: The sparsity masks are calibrated once before training, based on parameter sensitivity scores calculated on a representative data subset. The primary method used is Fisher Information, which estimates the importance of each parameter to the model's output [9]. Parameters with the lowest Fisher scores are considered less important and are set to 0 in the mask.

- Masking Strategies (Personal Contribution): To rigorously evaluate this approach, we systematically compared several masking strategies at a fixed sparsity ratio:

  1. Sensitivity-Based: The main strategy using Fisher Information to prune the least sensitive parameters.
  2. Heuristic: A Magnitude-based strategy that prunes parameters with the smallest absolute weights.
  3. Baseline: A Random masking strategy that prunes a random subset of parameters, serving as a simple baseline.
  4. Novelty: A Hybrid strategy was also tested, combining elements of the other methods.

3.4. Mask Overlap Analysis

To quantify the similarity between the sparsity masks generated for different clients and across different strategies, an overlap analysis was conducted. The metric used is the Jaccard Index, defined as the size of the intersection divided by the size of the union of the sets of active parameters of two masks, A and B:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

A high Jaccard Index value (close to 1) indicates that two clients deem the same subset of parameters as "important," while a low value (close to 0) suggests greater specialization and personalization of the model on the client.

The following section will present the results obtained from this comparison.

4. Experiments and results

This section describes the experimental setup adopted for the evaluation and presents the quantitative results obtained.

4.1. Experimental setup

- Implementation and Framework: All experiments were implemented in Python using the PyTorch framework. The Vision Transformer model was instantiated using the timm library. The experiments were run on Google Colab.

- Centralized Training Configuration: To establish a solid baseline, the centralized training was optimized via a grid search on key hyperparameters (learning rate, batch size, weight decay), with the aid of a cosine annealing learning rate scheduler and an Early Stopping strategy to mitigate overfitting.

- Federated Learning Configuration: The federated experiments simulated K=100 clients. In each communication round, a fraction C=0.1 (10 clients) was randomly selected. Each selected client performed local training for J=4, 8, 16 epochs before aggregation via FedAvg.

- Evaluation Metrics: The primary performance metric is top-1 test accuracy on the CIFAR-100 dataset. The Jaccard Index was used for the mask consistency analysis.

4.2. Centralized Baseline Performance

To establish a rigorous performance baseline, a grid search was conducted on the hyperparameters for the centralized training. Various combinations of learning rate (LR) and batch size (BS) were tested to identify the optimal configuration for the ViT-Small model on the CIFAR-100 dataset.

Experiment #	Hyperparameters	Max Accuracy (%)
1	LR=0.01, BS=64	88.3
2	**LR=0.01, BS=128**	**90.3**
3	LR=0.03, BS=64	82.3
4	LR=0.03, BS=128	86.2
5	LR=0.05, BS=64	77.5
6	LR=0.05, BS=128	83.4

Table 1. Grid Search results for centralized training

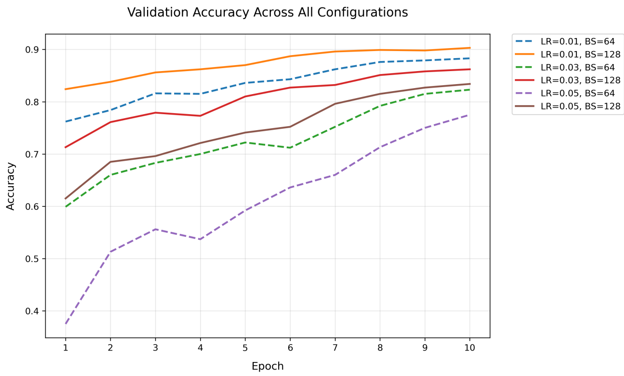

Figure 1. Validation accuracy curves for different hyperparameter configurations during centralized training.

Figure 1 shows the accuracy learning curves for the different configurations tested. A significant variation in performance can be noted, highlighting the importance of hyperparameter selection. The top curve, which shows the fastest convergence and the highest accuracy value, corresponds to the optimal configuration.

The quantitative analysis of the results, presented in Table 1, summarizes the performance of each experiment. The best configuration (Experiment #2), with a learning rate of 0.01 and a batch size of 128, achieved a maximum accuracy of 90.3%. This value will be used as the gold standard reference to evaluate the impact of decentralization and data heterogeneity in subsequent experiments.

4.3. Federated learning performance

After establishing the centralized baseline (90.3%), our analysis focused on evaluating the performance of the FedAvg algorithm in decentralized environments. We examined two key conditions: an ideal scenario with IID data and a series of realistic scenarios with non-IID data.

The first federated experiment, under IID conditions, was designed to isolate the intrinsic cost of the federation process. As shown in Figure 2, the model achieves a final validation accuracy of 70.5%. This drop of nearly 20 percentage points from the centralized baseline highlights a significant information loss due to the parameter averaging process, even in the absence of data heterogeneity.

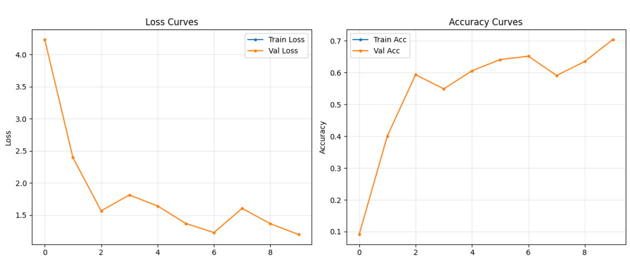

Figure 2. Validation accuracy and loss curves for the Federated Learning IID scenario.

Classes per client	Local Epochs	Accuracy (%)
1	**4**	**1.2**
1	8	1.0
1	16	1.0
5	4	74.1
5	8	41.0
5	16	6.3
10	4	83.6
10	8	79.5
10	16	52.7
50	**4**	**84.1**
50	8	79.2
50	16	68.1

Table 2. Final Test Accuracy (%) for Federated Non-IID Scenario.

Next, we introduced the challenge of statistical heterogeneity by analyzing several non-IID scenarios. The results, summarized in Table 2, reveal a complex dynamic. Surprisingly, under mild (Nc=50) and moderate (Nc=10) heterogeneity, the model achieves accuracies of 84.1% and 83.6%, respectively. These values, while lower than the centralized baseline, are unexpectedly higher than the IID result. This counter-intuitive finding suggests that the non-IID runs may have benefited from a more favorable hyperparameter configuration that was not present in the IID experiment, making a direct comparison between the two problematic.

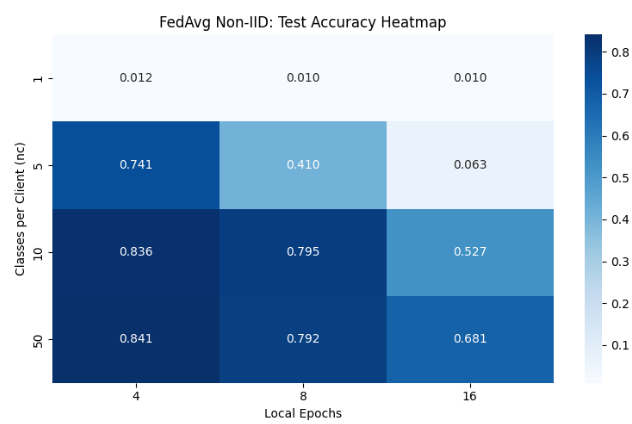

Figure 3. Heatmap summarizing the combined impact of Nc and J on performance.

However, the non-IID analysis confirms two fundamental hypotheses of FL. First, as expected, performance degrades dramatically as data skew increases: accuracy drops to 74.1% for Nc=5 and collapses to 1.2% in the extreme case of Nc=1. Second, the results highlight the risk of client drift: as seen in Figure 3, for any given Nc, increasing the number of local epochs (J) leads to a performance collapse (e.g., from 83.6% to 52.7% for Nc=10 as J goes from 4 to

16). This demonstrates that excessive local computation on heterogeneous data is counter-productive.

In summary, while the federation process itself introduces a significant performance cost, it is data heterogeneity, especially when combined with excessive local computation, that poses the greatest obstacle for standard FedAvg. This provides a strong motivation for exploring sparse training techniques as a potential solution.

4.4. Efficacy of Sparse Training as a Model Editing Technique

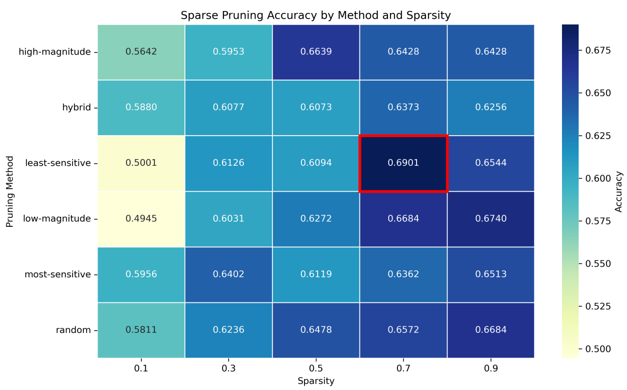

Figure 4. Test accuracy vs. sparsity ratio for the six different masking criteria.

Sparse training demonstrates strong potential for improving FL scalability. By applying gradient masks to restrict updates to only a fraction of the parameters, the system significantly reduces communication overhead and local computational costs. Among the methods tested, least-sensitive pruning—based on Fisher Information—achieves the most favorable trade-off between model accuracy and sparsity. At 70% sparsity, models maintain nearly 69% accuracy, demonstrating robustness against the reduction in parameter updates.

Other approaches, such as pruning the most sensitive or highest magnitude weights, led to steep performance declines. These strategies tend to remove core predictive features, which are essential in both centralized and federated learning environments. Random and low-magnitude masking showed moderate results but lacked consistency. The hybrid approach balanced stability and accuracy but did not surpass the Fisher-based method.

These findings validate that informed parameter selection is critical to effective sparse training. The results reinforce the idea that optimal sparse updates should preserve essential task-relevant information while promoting communication-efficient collaboration across clients.

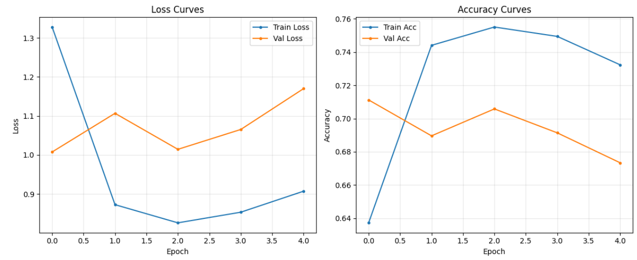

Figure 5. Accuracy and loss curves for the best sparse configuration (least-sensitive, 70% sparsity).

4.5. Analysis of Mask Overlap and Client Specialization

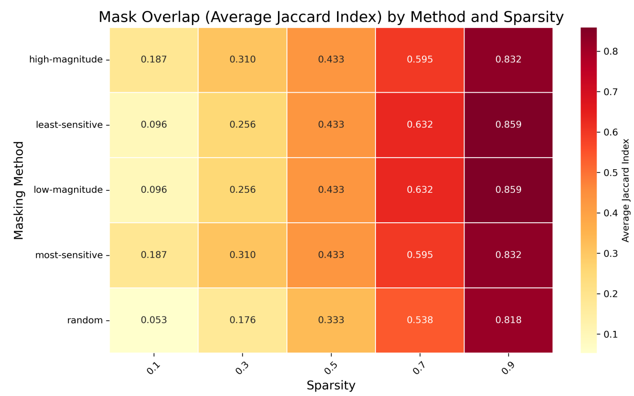

Figure 6. Heatmap of the average Jaccard Index, showing the overlap between client masks for different methods and sparsity ratios.

To investigate the role of mask diversity across clients, we measured the Jaccard Index of their masks. The analysis revealed that the least-sensitive strategy resulted in the lowest mask overlap between clients, indicating strong client specialization. This diversity allows each client to tailor updates to its local data distribution, helping the global model integrate heterogeneous patterns more effectively.

In contrast, the most-sensitive method produced high overlap across clients' masks. This uniformity exacerbates parameter interference during aggregation, especially under non-IID conditions, and ultimately degrades the model's ability to generalize. Similarly, the hybrid method maintained moderate overlap and intermediate performance.

The mask overlap metric thus provides a lens through which we can understand personalization in FL. Encouraging mask diversity enables more flexible learning, reduces contention over shared parameters, and allows clients to preserve their local characteristics without compromising global learning objectives. This contributes directly to mitigating client drift and enhancing convergence stability.

5. Discussion

The results of this study highlight the multifaceted challenges of Federated Learning (FL) in non-IID environments and the potential of model editing through sparse training to mitigate those challenges. The experiments demonstrate that data heterogeneity and excessive local updates can drastically impair model convergence and final accuracy. However, the introduction of sparsity not only helps reduce the computational burden on individual clients but also provides a mechanism for promoting personalization.

Least-sensitive masking, which leverages Fisher Information to identify non-critical parameters, emerges as the most effective approach. It enables each client to focus updates on a distinct parameter subset, thereby minimizing parameter interference during aggregation and preserving diverse local patterns. This finding aligns with the observed low Jaccard index values for least-sensitive masks, indicating low overlap and higher specialization across clients.

Moreover, the results show that sparsity does not inherently compromise learning capacity. When applied judiciously, sparse updates can retain significant portions of the centralized model's accuracy while substantially lowering communication costs—a key concern in real-world federated setups. The importance of intelligent mask calibration is evident, as naive strategies such as random or high-magnitude pruning degrade performance and consistency.

While promising, this work also reveals certain limitations. The use of static masks precludes dynamic adaptation during training, and experiments were conducted in a simulated FL setting using Colab resources, which may not reflect real-world deployment constraints. Additionally, the synthetic data splits—although effective for controlled analysis—do not fully capture the complexity of real user distributions.

Future Work:

- Implement dynamic sparsity mechanisms where masks evolve during training based on updated sensitivity scores.
- Explore adaptive sparsity levels tailored to client capabilities or data complexity.
- Investigate hybrid FL strategies combining sparse training with other personalization methods (e.g., meta-learning or client clustering).
- Extend evaluation to real-world federated datasets to test robustness across diverse settings.

6. Conclusions

This work presents a systematic exploration of sparse model editing in the context of Federated Learning using Vision Transformers. We show that combining FL with sparse fine-tuning, particularly through least-sensitive masking, yields compelling benefits: reduced communication and computation costs, improved client personalization, and competitive performance under non-IID conditions.

By analyzing centralized, IID, and non-IID FL scenarios, and applying multiple sparsity strategies, we demonstrate that sparsity-aware updates can substantially close the performance gap between centralized and decentralized learning. The Jaccard overlap analysis further supports the hypothesis that personalization is enhanced through selective parameter updates, offering a path to mitigate client drift and statistical heterogeneity.

Our modular framework and experimental insights lay the groundwork for future improvements in scalable, efficient, and privacy-conscious federated systems. With further development, sparse model editing may become a core component of next-generation FL deployments, enabling real-world applications across domains with distributed data and constrained resources.

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